Project: Worked examples
Title: Sealed heating system: emitters and expansion vessel
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A closed LTHW system below 100 °C: the catalogue output (at ΔT 50 K) each room's emitter needs at the design flow, return and room temperatures, checked against the selected emitters; and the minimum expansion vessel for the system volume, fill and maximum temperatures, static head and pressures.

WHAT IT CALCULATES	RELATION
Emitter temperature difference	$\Delta T = (t_f + t_r)/2 - t_i$, or the log mean $(t_f - t_r)/\ln((t_f - t_i)/(t_r - t_i))$
Output at design	$\Phi = \Phi 50 \cdot (\Delta T/50)^n$; required catalogue output $\Phi 50 = \Phi_{req}/(\Delta T/50)^n$
System volume	$V_s = \Sigma V_{em} + V_{pipe} + V_{plant}$
Water expansion	$e_x = \rho(t_{fill})/\rho(t_{max}) - 1$ (water density by Kell 1975), or a glycol record
Vessel pre-charge	$p_i = \rho \cdot g \cdot h_{static}$ + margin (≥ static head)
Vessel size	$V_e = e_x \cdot V_s/[1 - (p_i + 1)/(p_f + 1)]$ (pressures in bar gauge, + 1 bar for absolute); the expansion volume only, sized for the higher of t_{max} and t_f; a water reserve and a safety-valve margin are not included

Basis: Radiators and convectors — test methods and rating (characteristic equation)[1](cl. 5); Heating — sealed systems and expansion[2](2016, §1.4).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
rooms	Rooms and emitters	5 row(s)		spec: Room heat loss calculation (B4) and radiator survey	any
t_f	Flow temperature	55	°C	spec: Heat pump design flow temperature	30 … 90 °C
t_r	Return temperature	45	°C	spec: Heat pump design return temperature	30 … 90 °C
t_i	Room temperature	21	°C	spec: Living-room design temperature	16 … 24 °C
n	Emitter exponent	1.3		assumption: Panel radiators, manufacturer exponent about 1.3.	1.1 … 1.5

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dT_{method}	Mean temperature difference	arithmetic mean (catalogue rating)		assumption: Catalogue ratings use the arithmetic mean.	any
V_{pipe}	Pipework water content	40	L	assumption: 15 mm and 22 mm copper, about 120 m.	0 … 10000 L
V_{plant}	Plant water content (boiler, cylinder coil, buffer)	15	L	spec: Heat pump and plate exchanger content, datasheet	0 … 10000 L
t_{fill}	Fill temperature	10	°C	assumption: Mains water at fill.	4 … 30 °C
t_{max}	Maximum system temperature	60	°C	spec: Heat pump maximum flow temperature	30 … 99 °C
h_{static}	Static height above the vessel	7	m	drawing: M-SEC-001 rev A	0 … 50 m
p_{margin}	Pre-charge margin over static head	0.3	bar	assumption: Usual margin over static head.	0 … 1 bar
p_f	Final (maximum working) pressure, gauge	2.5	bar	assumption: 0.5 bar below the safety valve.	0.5 … 6 bar
p_{prv}	Safety valve setting, gauge	3	bar	spec: Heat pump safety valve	1.5 … 6 bar
V_{vessel}	Selected vessel volume	12	L	spec: Site survey: existing 12 L vessel	1 … 5000 L

TABLE ROW	VALUES
rooms[1]	name = Lounge; Phi = 1800 W; Phi50 = 3600 W; V_em = 12 L
rooms[2]	name = Kitchen; Phi = 1100 W; Phi50 = 2200 W; V_em = 8 L
rooms[3]	name = Bed 1; Phi = 900 W; Phi50 = 1800 W; V_em = 7 L
rooms[4]	name = Bed 2; Phi = 700 W; Phi50 = 1400 W; V_em = 5 L
rooms[5]	name = Bathroom; Phi = 600 W; Phi50 = 1100 W; V_em = 4 L

3. Assumptions

A1. Closed LTHW system below 100 °C; emitter output follows Φ50·(ΔT/50)^n with the manufacturer's n. (Catalogue ratings at 75/65/20 °C.)
A2. Water density from the Kell (1975) correlation, within about 0.01 % of IAPWS-95 at atmospheric pressure, 0–100 °C; glycol mixtures take e from a record. (Formula, not a table.)
A3. Gauge pressures are made absolute by adding 1 bar, as the vessel relation is usually written. (Within 1.3 % of the standard atmosphere.)
A4. V_e is the expansion volume by CIBSE Guide B1's relation only. It does not include a water reserve for system losses, nor a margin between the final pressure and the safety valve setting beyond p_f below

p_{prv}; add both when the vessel is selected, from the heating-system standard in use. (Stated omission.)
A5. n (Emitter exponent) is taken as 1.3. (Panel radiators, manufacturer exponent about 1.3.)
A6. dT_{method} (Mean temperature difference) is taken as arithmetic mean (catalogue rating). (Catalogue ratings use the arithmetic mean.)
A7. V_{pipe} (Pipework water content) is taken as 40 L. (15 mm and 22 mm copper, about 120 m.)
A8. t_{fill} (Fill temperature) is taken as 10 degC. (Mains water at fill.)
A9. p_{margin} (Pre-charge margin over static head) is taken as 0.3 bar. (Usual margin over static head.)
A10. p_f (Final (maximum working) pressure, gauge) is taken as 2.5 bar. (0.5 bar below the safety valve.)

4. Working

[14] $dT = (t_f + t_r) / 2 - t_i$ [1](cl. 5)
$= (55\ degC + 45\ degC) / 2 - 21\ degC$
$= 29\ K$
Mean water-to-room temperature difference

[15] $dT_{fr} = t_f - t_r$
$= 55\ degC - 45\ degC$
$= 10\ K$
Flow-to-return temperature difference (what the water gives up across the emitters)

[16] $t_m = (t_f + t_r) / 2$
$= (55\ degC + 45\ degC) / 2$
$= 50\ degC$
Mean water temperature, (flow + return)/2

[17] $f_{dT} = (dT / 50\ K)^n$
$= (29\ K / 50\ K)^1.3$
$= 0.49256$
Output correction from ΔT 50 K

[18] $R1 = \Phi_{req}/f_{dT}$
$= 1800 / 0.49256$
$= 3654.4\ W$
Lounge: catalogue output needed; selected 3600 W gives 1773.2 W (utilisation 1.02)

[19] $R2 = \Phi_{req}/f_{dT}$
$= 1100 / 0.49256$
$= 2233.2\ W$
Kitchen: catalogue output needed; selected 2200 W gives 1083.6 W (utilisation 1.02)

[20] $R3 = \Phi_{req}/f_{dT}$
$= 900 / 0.49256$
$= 1827.2\ W$
Bed 1: catalogue output needed; selected 1800 W gives 886.6 W (utilisation 1.02)

[21] $R4 = \Phi_{req}/f_{dT}$
$= 700 / 0.49256$
$= 1421.2\ W$
Bed 2: catalogue output needed; selected 1400 W gives 689.58 W (utilisation 1.02)

[22] $R5 = \Phi_{req}/f_{dT}$
$= 600 / 0.49256$
$= 1218.1\ W$
Bathroom: catalogue output needed; selected 1100 W gives 541.81 W (utilisation 1.11)

[23] $\Phi_{tot} = \Sigma\ \Phi_{req}$
$= 5100\ W$
$= 5100\ W$
Total heat required by the rooms

[24] $V_{em} = \Sigma\ V_{em}$
$= 36\ L$
$= 36\ L$
Emitter water content

[25] $V_s = V_{em} + V_{pipe} + V_{plant}$
= 36 L + 40 L + 15 L
= 91 L
System water volume
[26] $\rho_{fill} = \rho(t_{fill})$, Kell 1975
= t = 10 °C
= 999.7 kg/m³
Water density at fill
[27] $\rho_{max} = \rho(t_{max})$, Kell 1975
= t = 60 °C
= 983.2 kg/m³
Water density at maximum temperature
[28] $e_x = \rho_{fill} / \rho_{max} - 1$
= 999.7 kg/m³ / 983.2 kg/m³ − 1
= 0.016783
Expansion coefficient of water
[29] $p_{st} = \rho_{fill} * 9.80665\ m/s^2 * h_{static}$
= 999.7 kg/m³ × 9.80665 m/s^2 × 7 m
= 0.68626 bar
Static head at the vessel
[30] $p_i = p_{st} + p_{margin}$
= 0.68626 bar + 0.3 bar
= 0.98626 bar
Pre-charge (initial) pressure, gauge
[31] $V_e = e_x * V_s / (1 - (p_i + 1\ bar) / (p_f + 1\ bar))$ [2](§1.4)
= 0.016783 × 91 L / (1 − (0.98626 bar + 1 bar) / (2.5 bar + 1 bar))
= 3.5312 L
Minimum expansion vessel volume
[32] $u_{emit} = \max$ over rooms $\Phi_{req}/(\Phi 50 \cdot f_{dT})$
= worst room
= 1.1074
Emitter utilisation (governing room)
[34] $u_{vessel} = V_e / V_{vessel}$
= 3.5312 L / 12 L
= 0.29427
Vessel utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
dT	29 K	Mean temperature difference
dT_{fr}	10 K	Flow-to-return temperature difference
t_m	50 degC	Mean water temperature
f_{dT}	0.49256	Output correction factor
R1	3654.4 W	Lounge: Φ50 needed
R2	2233.2 W	Kitchen: Φ50 needed
R3	1827.2 W	Bed 1: Φ50 needed
R4	1421.2 W	Bed 2: Φ50 needed
R5	1218.1 W	Bathroom: Φ50 needed
Φ_{tot}	5100 W	Total heat required

SYMBOL	VALUE	DESCRIPTION
V_{em}	36 L	Emitter water content
V_s	91 L	System water volume
e_x	0.016783	Expansion coefficient
p_i	0.98626 bar	Pre-charge pressure
V_e	3.5312 L	Minimum vessel volume
u_{emit}	1.1074	Emitter utilisation
u_{vessel}	0.29427	Vessel utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Selected emitters against room heat (governing room)	u_{emit} = 1.1074	u_{emit} ≤ 1	1.11	
PASS	Selected vessel against minimum volume	u_{vessel} = 0.29427	u_{vessel} ≤ 1	0.294	[2](§1.4)

7. Applicability

WARNING — record.n: n: a tabulated value not taken from a reference record (Emitter exponent); enter it as one so its source, edition and licence travel with it.

8. Verification

```
Template bs.sealed-system@1.2.1 Content hash ae2947ba87fac092 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. BS EN 442-2:2014.
2. CIBSE Guide B1.